

PS Code	Problem Statements
HPC1	Design and implement Parallel Breadth First Search and Depth First Search based on existing algorithms using OpenMP. Use a Tree or an undirected graph for BFS and DFS .
HPC2	Write a program to implement Parallel Bubble Sort and Merge sort using OpenMP. Use existing algorithms and measure the performance of sequential and parallel algorithms.
HPC3	Implement Min, Max, Sum and Average operations using Parallel Reduction.
HPC4	Write a CUDA Program for : Addition of two large vectors
HPC5	Write a CUDA Program for : Matrix Multiplication using CUDA C
HPC6	Implement HPC application for AI/ML domain.
DL1	Linear regression by using Deep Neural network: Implement Boston housing price prediction problem by Linear regression using Deep Neural network. Use Boston House price prediction dataset.
DL2	Classification using Deep neural network: Multiclass classification using Deep Neural Networks: Example: Use the OCR letter recognition dataset https://archive.ics.uci.edu/ml/datasets/letter+recognition
DL3	Classification using Deep neural network: Binary classification using Deep Neural Networks Example: Classify movie reviews into positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset
DL4	Convolutional neural network (CNN)) : Use any dataset of plant disease and design a plant disease detection system using CNN.
DL5	Convolutional neural network (CNN): Use MNIST Fashion Dataset and create a classifier to classify fashion clothing into categories.
DL6	Recurrent neural network (RNN) Use the Google stock prices dataset and design a time series analysis and prediction system using RNN.